

Highlights

Active Contact Localization Without Joint-Torque or Force/Torque Sensing: Ambiguity-Preserving Inference for Robot Manipulators

- Contact location and force are measured from joint states and commanded torques only.
- A single view determines a line; two transverse views provably determine the point.
- Closed-form information bound links accuracy to force magnitude and view separation.
- Line-of-action support preservation suppresses confident-but-wrong estimates.
- A GUM-type residual uncertainty budget quantifies the sensorless measurement.

Active Contact Localization Without Joint-Torque or Force/Torque Sensing: Ambiguity-Preserving Inference for Robot Manipulators

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ABSTRACT

We measure the location of an external contact on an unknown grasped object, together with the contact force, from the joint positions, velocities, and commanded torques of a 7-DoF manipulator, without joint-torque, force/torque, or tactile sensors or an object model. The difficulty is structural rather than stochastic: from a single arm configuration, every point on the force line of action produces the same joint residual. We prove that two orientations with transverse object-frame force directions identify the contact point uniquely, and derive a closed-form information bound that scales with the force magnitude and the sine of half the inter-view force angle. The proposed method validates the payload-compensated momentum-observer residual with a frozen trust gate, expresses its uncertainty with a GUM-type budget that includes follower-force variation, represents the ambiguity with a force-profiled particle filter whose moves and re-supply steps all target the same posterior, keeps hypotheses alive along the unresolved line of action, and selects the next orientation by profile Fisher information inside the validated envelope. In 1,200 paired trials on a simulated Franka Research 3 with realistic model and actuation errors, the method reduces the 95th-percentile localization error from ?? mm for a regularized particle filter to 21.9 mm and the confident-but-wrong rate from ??% to 0.8%, at median position and force-magnitude errors of 6.6 mm and 1.3%; after two views, the actively selected orientation reduces the 95th-percentile error from 40.4 to 25.9 mm relative to a random one. **On the real robot, the method localizes the equivalent point force of calibrated added masses with a median error of HWMED mm and pressed contacts with HWHCMED mm, and its reported 95% radius contains the reference in HWBALLALL% of the 144 trials.** Project page: <https://anonymous.4open.science/w/apcl/>.

1. Introduction

A manipulator that knows where and how hard it is being touched can react safely to people, comply with a guiding hand, and reason about the objects it holds. Measuring this contact information normally requires dedicated instrumentation: joint-torque sensors [1], a wrist force/torque (F/T) sensor [2], or tactile skins. Many industrial and low-cost arms carry none of these, and a grasped tool or object extends the contact surface beyond any skin that the arm might carry. This paper treats contact localization as an indirect measurement problem: the joint encoders and the commanded torque already available in every torque-controlled arm constitute the measurement chain, and the task is to extract the contact point on an unknown grasped object and the contact force from them, together with an uncertainty statement that can be trusted [3].

Momentum observers recover the external joint torque from proprioceptive signals without acceleration measurement [1, 4, 5], data-driven detectors flag collision events from internal robot signals [6], and contact particle filters turn observer residuals into contact-location posteriors [7, 8]. The key obstacle to localizing a contact on an unknown object, however, is not observer noise. A single contact force $\mathbf{f}$ applied at point $\mathbf{p}$ produces the moment $\mathbf{p} \times \mathbf{f}$, and every point $\mathbf{p} + \alpha \mathbf{f}$ produces exactly the same moment. From a single arm configuration, the joint residual therefore constrains the contact to a line, whatever the number of joints, the observer bandwidth, or the sensor resolution. This line-of-action ambiguity is classical in intrinsic contact sensing [2, 9], where it is removed by assuming a known surface. With an unknown object, no surface is available to intersect the line.

We remove the ambiguity by changing the measurement configuration instead of adding a sensor. Rotating the grasped object about its current position changes the object-frame force direction while the same material point stays in contact, so a second view yields a second line that intersects the first at the true contact. Exploiting this idea with a finite-sample estimator raises a reliability issue that we identify as the dominant failure mode. Between views, the posterior is legitimately line-shaped, but the particles that lie on the line are few, and resampling concentrates them on a random stretch of it. The particles near the true contact may disappear before the second view arrives, and the filter then converges to a narrow posterior that excludes the truth: a *confident-but-wrong* (CBW) measurement. A narrow posterior is useful only if it contains the true contact.

Our approach is summarized as *preserve first, disambiguate actively*. The contributions are:

1. **Measurement observability.** We prove that the single-view contact measurement is non-identifiable exactly along the object-frame force direction, that two views identify the contact point and the world-frame force uniquely if and only if their object-frame force directions are not parallel, and we give a closed-form Fisher-information bound in terms of force magnitude and inter-view angle (Section 3.2).
2. **Calibrated sensorless residual.** We combine a compact payload calibration and a frozen trust gate with a GUM-type uncertainty budget whose Type A part is evaluated from held-out calibration residuals and whose Type B part covers friction, torque-constant tolerance, and follower-force variation (Section 3.3).
3. **Ambiguity-preserving inference.** We introduce a force-profiled contact particle filter in which every step targets the same posterior: Metropolis–Hastings moves that leave the posterior invariant keep hypotheses on the likelihood ridge, and a recovery layer that distinguishes weight degeneracy, model inconsistency, and loss of support along the unresolved line of action re-supplies hypotheses by importance sampling with target-over-proposal weights, along the theoretical line of action whenever the accumulated information along it is insufficient (Sections 3.4 and 3.5).
4. **Active measurement design.** We select the next orientation inside the validated envelope by maximizing the posterior-expected profile Fisher information of the contact point, with the force as a nuisance parameter, under a position-preserving motion that keeps the follower contact intact (Section 3.6).

We quantify accuracy, tail error, CBW rate, and credible-region coverage in 1,200 paired simulation trials, and show that support preservation is what makes the recursive measurement reliable and active sensing what makes it efficient (Section 4). A physical-contact study then establishes what the pusher must provide for the measurement to hold: tangential compliance (Section 4.7). An added-mass protocol with a traceable reference then validates the measurement chain without any contact mechanics and separates the precision of the sensorless measurement from its trueness (Section 4.8).

2. Related work

External torque estimation Generalized-momentum observers [4, 5] estimate the external joint torque as a first-order filtered residual and underpin collision detection, isolation, and identification [1]. Cartesian contact wrenches have been estimated from motor currents [10], from the observer residual under an assumed contact location [11, 12], and from identified dynamics on hydraulic manipulators [13]. Unmodeled dynamics and actuator faults contaminate such residuals and can be learned with dedicated identification [14]; friction in particular is a dominant residual component [15]. These methods measure the wrench when the contact location is known; they do not locate the contact on an unknown object.

Contact localization Intrinsic contact sensing recovers the contact point from a wrist F/T measurement and a known surface [2, 9]. On the robot body, the contact link and point can be isolated from the residual [1, 16], sensing redundancy improves localization on torque-sensing robots [17], and the contact particle filter [7] localizes contacts on a known robot mesh from proprioception. Fed with the generalized-momentum residual computed from motor currents and encoders, the contact particle filter also localizes contacts on a collaborative robot without joint-torque sensors [8]. Learning-based localization maps proprioceptive signals to contact points [18], including on arms in motion without joint-torque measurements [19]. All of these rely on known geometry, which removes the line-of-action ambiguity by intersection. Even then, distinct contacts on the surface can produce nearly identical joint torques, so the attainable estimate is a set of candidate points, whose spurious members are falsified by small exploratory motions [20]. We address an unknown grasped object, where the ambiguity must be represented and resolved.

Payload identification The inertial parameters of a load are linear in the joint-space dynamics [21] and can be identified online for unknown held objects [22], or directly with wrist or fingertip F/T sensors [23]. Regularization and physical-consistency constraints keep the identified parameters feasible under limited excitation [24]. We need only mass and center of mass and identify them from a compact, contact-free orientation sweep.

Particle degeneracy Sample impoverishment in particle filters is usually monitored by the effective sample size and countered by resampling and kernel roughening [25, 26], or, for static parameters, by Markov chain Monte Carlo moves that leave the posterior invariant [27]. These criteria detect a collapse of the weights, but not a loss of support

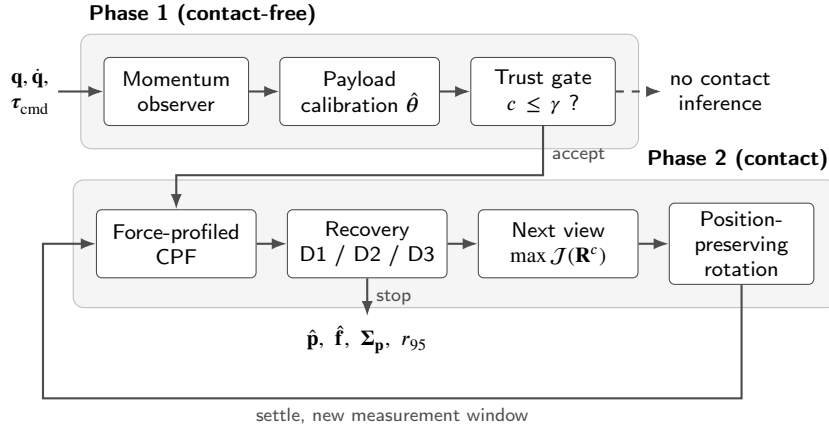


Figure 1: Sensorless contact measurement pipeline. Phase 1 is contact-free; Phase 2 alternates measurement windows, inference, and reorientation.

along a direction that the current measurement cannot resolve; our results show that the latter dominates the tail error in contact localization.

Active sensing and experiment design Information-based selection of measurement configurations is standard in optimal experiment design [28] and in robot identification [29], and interactive perception exploits deliberate motion to reveal hidden properties [30]. Tactile localization with particle filters has used information gain to choose touches [31]. In visually occluded scenes, contact-rich rummaging plans robot trajectories online by maximizing their mutual information with a particle belief over the object pose [32]. Unlike exploratory motions that falsify a finite set of candidate contacts on a known surface [20], our criterion scores each reorientation against the continuous line-of-action posterior of an unknown object. We apply the same principle to proprioceptive contact localization, with a profile information measure that eliminates the unknown force.

3. Sensorless contact measurement

This section defines the measurement model, establishes what it can and cannot identify, and derives the measurement method from this analysis. Fig. 1 summarizes the method. Phase 1 runs before contact and delivers a payload-compensated residual that has passed a model-validity test, together with its uncertainty budget (Section 3.3). Phase 2 runs after contact onset and alternates quasi-static measurement windows, particle updates with recovery, and information-driven reorientation, without any reset of the robot or observer state between phases (Sections 3.4–3.6).

3.1. Measurement model

A fixed-base 7-DoF manipulator holds an unknown object rigidly. Let $\{E\}$ be a frame attached to the end effector and the object, and let $\mathbf{R}_k \in SO(3)$ and $\mathbf{o}_k \in \mathbb{R}^3$ be its orientation and position in the world frame $\{W\}$ at measurement pose k . The available measurements and the unavailable ones are listed in Table 1. The unknowns are the contact point $\mathbf{p} \in \mathcal{B} \subset \mathbb{R}^3$ expressed in $\{E\}$, where $\mathcal{B}$ is a bounded search region around the end effector, and the world-frame contact force $\mathbf{f} \in \mathbb{R}^3$.

We consider the following contact regime. (A1) A single contact acts at one material point $\mathbf{p}$ of the object throughout an episode. (A2) The world-frame force $\mathbf{f}$ is constant over the episode up to a small follower variation, which is modeled explicitly in Section 3.3. (A3) The contact is a follower contact, such as a human hand or a force-controlled pusher that is compliant tangentially to the surface, which stays on the same material point while the object is reoriented. (A4) Measurements used for localization are taken in quasi-static windows after each reorientation has settled.

Residual model The generalized-momentum observer [4, 5]

$$\mathbf{r}(t) = \mathbf{K}_O \left(\mathbf{M}(\mathbf{q})\dot{\mathbf{q}} - \int_0^t (\boldsymbol{\tau}_{\text{cmd}} + \mathbf{C}^\top(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} - \mathbf{g}(\mathbf{q}) - \mathbf{D}\dot{\mathbf{q}} + \mathbf{r}) ds \right), \quad (1)$$

Table 1

Measurement availability.

Signal	Used by estimator	Role
Joint position q , velocity $\dot{q}$	Yes	Input
Commanded joint torque τ_{cmd}	Yes	Input
Measured joint torque	No	—
Wrist F/T sensor	No	—
Tactile sensing	No	—
Object geometry	No	—
Contact point and force	No	Ground truth only

where $\mathbf{D}$ is the nominal viscous joint damping, satisfies $\dot{\mathbf{r}} = \mathbf{K}_O(\tau_{\text{dist}} - \mathbf{r})$, i.e., $\mathbf{r}$ is a first-order low-pass estimate of the total disturbance torque. Evaluated with the nominal model of the arm, this disturbance contains the external contact torque, the gravity torque of the unknown payload, model and actuation errors, and noise:

$$\mathbf{r} = \tau_{\text{ext}} + \mathbf{Y}(\mathbf{q})\boldsymbol{\theta} + \boldsymbol{\delta} + \boldsymbol{\epsilon}. \quad (2)$$

The payload term is compensated in Phase 1, and the error terms are screened by the trust gate and quantified by the uncertainty budget (Section 3.3).

Let $\mathbf{J}_k \in \mathbb{R}^{6 \times 7}$ be the body Jacobian at pose k that maps joint velocities to the twist of $\{E\}$ expressed in $\{E\}$. The contact force expressed in $\{E\}$ is $\mathbf{f}_k = \mathbf{R}_k^\top \mathbf{f}$, and the equivalent wrench at the origin of $\{E\}$ is $\mathbf{w}_k = [\mathbf{f}_k^\top, (\mathbf{p} \times \mathbf{f}_k)^\top]^\top$. Hence

$$\tau_{\text{ext},k} = \mathbf{J}_k^\top \mathbf{w}_k = \underbrace{\mathbf{J}_k^\top \begin{bmatrix} \mathbf{I}_3 \\ [\mathbf{p}]_\times \end{bmatrix} \mathbf{R}_k^\top}_{\mathbf{G}_k(\mathbf{p})} \mathbf{f}, \quad (3)$$

where $[\cdot]_\times$ is the skew-symmetric matrix of the cross product. After payload compensation, the measurement model is $\mathbf{r}_k = \mathbf{G}_k(\mathbf{p})\mathbf{f} + \boldsymbol{\epsilon}_k$. It is linear in the force and bilinear in the contact point and force.

3.2. Observability

Proposition 1 (Single-view line-of-action ambiguity). *For any pose k , any $(\mathbf{p}, \mathbf{f})$, and any $\alpha \in \mathbb{R}$, $\mathbf{G}_k(\mathbf{p} + \alpha \mathbf{f}_k) \mathbf{f} = \mathbf{G}_k(\mathbf{p}) \mathbf{f}$. Consequently, the single-view likelihood is constant along the line $\{\mathbf{p} + \alpha \mathbf{f}_k\}$, and no estimator can resolve the contact point along $\mathbf{f}_k$ from pose k alone.*

PROOF. $[\mathbf{p} + \alpha \mathbf{f}_k]_\times \mathbf{f}_k = \mathbf{p} \times \mathbf{f}_k + \alpha \mathbf{f}_k \times \mathbf{f}_k = \mathbf{p} \times \mathbf{f}_k$.

Proposition 1 holds for any number of joints and any noise level. It carries over to the profile likelihood in which the force is estimated with a fixed noise covariance: if $\hat{\mathbf{p}}$ maximizes the single-view profile likelihood and $\hat{\mathbf{f}}$ is its profiled force, every point $\hat{\mathbf{p}} + \alpha \hat{\mathbf{f}}_k$ attains the same maximal value, because $\hat{\mathbf{f}}$ reproduces the same fitted residual there. The single-view profile likelihood thus has a flat ridge along a whole line instead of a peak. A line-shaped single-view posterior is therefore not an estimator failure; it is the correct representation of the single-view inverse problem.

Proposition 2 (Two-view identifiability). *Let $\mathbf{J}_1$ and $\mathbf{J}_2$ have full row rank. The noise-free measurements $\mathbf{G}_1(\mathbf{p})\mathbf{f}$ and $\mathbf{G}_2(\mathbf{p})\mathbf{f}$ determine $(\mathbf{p}, \mathbf{f})$ with $\mathbf{f} \neq \mathbf{0}$ uniquely if and only if $\mathbf{f}_1 \times \mathbf{f}_2 \neq \mathbf{0}$. Let the relative rotation $\Delta \mathbf{R} = \mathbf{R}_2 \mathbf{R}_1^\top$ be a rotation by the angle $\varphi \in (0, \pi]$ about the unit axis $\mathbf{a}$. Then the condition holds if and only if $\mathbf{a}$ is not parallel to $\mathbf{f}$ and $\Delta \mathbf{R}$ is not a half-turn about an axis perpendicular to $\mathbf{f}$.*

PROOF. Full row rank of $\mathbf{J}_k$ makes $\mathbf{J}_k^\top$ injective, so each view determines the wrench $\mathbf{w}_k$. Suppose $(\mathbf{p}, \mathbf{f})$ and $(\mathbf{p}', \mathbf{f}')$ give identical wrenches. The force components give $\mathbf{R}_k^\top \mathbf{f} = \mathbf{R}_k^\top \mathbf{f}'$, hence $\mathbf{f} = \mathbf{f}'$. The moment components give $(\mathbf{p} - \mathbf{p}') \times \mathbf{f}_k = \mathbf{0}$ for $k = 1, 2$, hence $\mathbf{p} - \mathbf{p}' \in \text{span}(\mathbf{f}_1) \cap \text{span}(\mathbf{f}_2)$, which is $\{\mathbf{0}\}$ if and only if $\mathbf{f}_1 \times \mathbf{f}_2 \neq \mathbf{0}$. If $\mathbf{f}_1 \parallel \mathbf{f}_2$, then $(\mathbf{p} + \alpha \mathbf{f}_1, \mathbf{f})$ reproduces both measurements for every α . Finally, $\mathbf{f}_1 \parallel \mathbf{f}_2$ if and only if $\Delta \mathbf{R}^\top \mathbf{f} \parallel \mathbf{f}$, i.e., $\mathbf{f}$ is an eigenvector of $\Delta \mathbf{R}$, which occurs exactly when $\mathbf{a} \parallel \mathbf{f}$ or when $\varphi = \pi$ with $\mathbf{a} \perp \mathbf{f}$.

Remark 1 (Per-view forces). The proof uses the force components only to fix the object-frame force of each view. Proposition 2 therefore also holds, with the condition $\mathbf{f}_1 \times \mathbf{f}_2 \neq \mathbf{0}$, when every view has its own unknown force $\mathbf{f}_k$ and only the contact point is shared: each view determines its own line of action, and two non-parallel lines intersect at the contact. Assumption (A2) is thus not what makes the contact point identifiable; it lets the estimator pool the force information of all views, which tightens the estimate.

Rotating the object about the force line therefore conveys no information, whereas any rotation with a component perpendicular to the force does. The payload calibration of Section 3.3 is the same problem with gravity as a known-direction force acting at the unknown center of mass, and Proposition 2 explains why it requires orientation diversity about horizontal axes.

The following bound quantifies how the accuracy grows with view separation. Consider the moment channel with isotropic noise $\sigma_m^2 \mathbf{I}_3$ and the force treated as known. The Fisher information on $\mathbf{p}$ from K views is

$$\mathbf{I}_p = \frac{1}{\sigma_m^2} \sum_{k=1}^K [\mathbf{f}_k]_{\times}^{\top} [\mathbf{f}_k]_{\times} = \frac{1}{\sigma_m^2} \sum_{k=1}^K (\|\mathbf{f}\|^2 \mathbf{I}_3 - \mathbf{f}_k \mathbf{f}_k^{\top}). \quad (4)$$

Corollary 1 (Two-view information bound). Let $K = 2$, and let $\vartheta \in [0, \pi/2]$ be the acute angle between the two lines of action, $\cos \vartheta = |\hat{\mathbf{f}}_1^{\top} \hat{\mathbf{f}}_2|$. The eigenvalues of $\mathbf{I}_p$ are $\|\mathbf{f}\|^2(1 \pm \cos \vartheta)/\sigma_m^2$ and $2\|\mathbf{f}\|^2/\sigma_m^2$, and the worst-direction Cramér–Rao bound on the contact-point standard deviation is

$$\sigma_p^{\text{worst}} \geq \frac{\sigma_m}{\sqrt{2} \|\mathbf{f}\| \sin(\vartheta/2)}, \quad \cos \vartheta = \left| \cos \varphi + (1 - \cos \varphi)(\mathbf{a}^{\top} \hat{\mathbf{f}})^2 \right|. \quad (5)$$

PROOF. The matrix $\hat{\mathbf{f}}_1 \hat{\mathbf{f}}_1^{\top} + \hat{\mathbf{f}}_2 \hat{\mathbf{f}}_2^{\top}$ has eigenvalues $1 \pm \hat{\mathbf{f}}_1^{\top} \hat{\mathbf{f}}_2$ and 0. Substituting into (4) gives the eigenvalues $\|\mathbf{f}\|^2(1 \mp \hat{\mathbf{f}}_1^{\top} \hat{\mathbf{f}}_2)/\sigma_m^2$ and $2\|\mathbf{f}\|^2/\sigma_m^2$, whose minimum is $\|\mathbf{f}\|^2(1 - \cos \vartheta)/\sigma_m^2 = 2\|\mathbf{f}\|^2 \sin^2(\vartheta/2)/\sigma_m^2$ for the acute angle ϑ . The angle relation follows from $\mathbf{f}_1^{\top} \mathbf{f}_2 = \mathbf{f}^{\top} \Delta \mathbf{R}^{\top} \mathbf{f}$ and Rodrigues' formula.

Because a line of action has no direction, the bound depends only on the acute angle between the two lines: it is inversely proportional to the force magnitude and to $\sin(\vartheta/2)$, and for a rotation about an axis perpendicular to the force, ϑ equals the rotation angle φ up to 90° and $\pi - \varphi$ beyond. In the estimator, the force is unknown and acts as a nuisance parameter; Section 3.6 uses the corresponding profile information, which accounts for the force uncertainty in the same way that precision bounds account for unknown systematic effects [33]. Fig. 2 compares the bound with the simulated two-view errors of Section 4.

3.3. Validated residual and uncertainty budget

Payload calibration Under (A4), inertial effects of the payload are negligible in the measurement windows, and its contribution to (2) is the gravity torque

$$\mathbf{Y}(\mathbf{q})\boldsymbol{\theta} = \mathbf{J}^{\top} \begin{bmatrix} \mathbf{R}^{\top} \mathbf{g}_W & \mathbf{0} \\ \mathbf{0} & -[\mathbf{R}^{\top} \mathbf{g}_W]_{\times} \end{bmatrix} \begin{bmatrix} m \\ m\mathbf{c} \end{bmatrix}, \quad \boldsymbol{\theta} = [m, mc_x, mc_y, mc_z]^{\top}, \quad (6)$$

which is linear in the four parameters [21, 22]. Restricting the identification to these four parameters is consistent with the quasi-static measurement protocol; the six inertia parameters are not excited by the measurement windows and are therefore neither needed nor identifiable from them. We estimate $\boldsymbol{\theta}$ by ridge-regularized least squares from $N_c = 7$ contact-free orientations: the initial orientation and six tilts of 20° about the horizontal axes $\pm \mathbf{e}_x$, $\pm \mathbf{e}_y$, and $\pm(\mathbf{e}_x + \mathbf{e}_y)/\sqrt{2}$, with the end-effector position held fixed. The compact sweep avoids large workspace motions that would disturb the grasp, and by Proposition 2 its tilts about two horizontal directions make $\boldsymbol{\theta}$ identifiable. Each orientation is reached in 1.5 s, allowed to settle for 0.5 s, and measured over a 1.0 s window.

Trust gate A small calibration residual shows that the regression interpolates the calibration poses, not that the model is valid: the four parameters can absorb a structured model error on a compact pose set. We therefore test the frozen calibration on $N_h = 3$ challenge orientations tilted by 35° about axes that are not used for fitting, with the statistic

$$c = \max_{j \in \mathcal{C}} \frac{1}{n} \left\| \frac{\mathbf{r}_j - \mathbf{Y}_j \hat{\boldsymbol{\theta}}}{\sigma_g} \right\|^2, \quad (7)$$

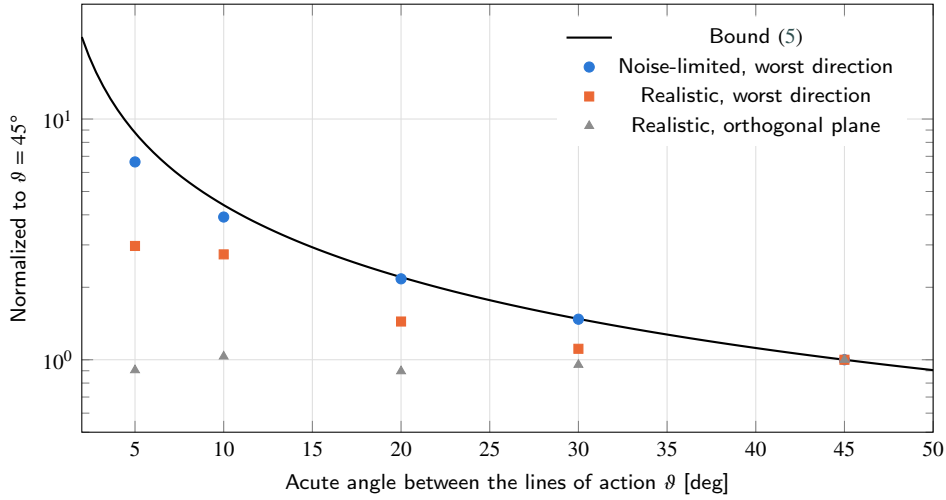


Figure 2: Two-view Cramér–Rao bound on the worst-direction contact-point standard deviation versus the acute angle between the lines of action, and the simulated RMS two-view errors along the worst direction and in the orthogonal plane, all normalized to $\theta = 45^\circ$. The noise-limited study follows the bound; under realistic errors, the worst-direction error still rises as the lines approach each other, while the orthogonal error stays at the bias floor.

Table 2

Residual uncertainty budget for joint i , Eq. (8).

Component	Type	Evaluation
Sub-window noise s_i^2/n	A	Pooled variance of 0.1-s means in calibration windows
Model error $\max(\sigma_i^{\text{LOO}}, \sigma_i^{\text{ch}})$	A	Leave-one-out and challenge residual RMS
Residual dry friction $a_i/\sqrt{3}$	B	Uniform, $a_i = 0.06 / 0.03$ N m (joints 1–4 / 5–7)
Torque constant $u_\kappa \bar{r}_i$	B	Relative tolerance $u_\kappa = 1\%$
Follower-force variation $\varepsilon_f \ \mathbf{f}\ $	B	Per view, isotropic, $\varepsilon_f = 3\%$, Eq. (9)

where $\sigma_g = 0.1$ N m is a fixed absolute scale and $n = 7$, and admit Phase 2 only if $c \leq \gamma$. The threshold γ is chosen once on a derivation set with offline validity labels, by the lexicographic rule: zero admitted CBW localizations, then zero admitted invalid models, then maximal acceptance of valid models. It is then frozen and evaluated on a disjoint held-out set. If the gate rejects, contact localization is not started. The challenge orientations also define the validated envelope within which Phase 2 is allowed to move.

Residual uncertainty budget A sensorless estimate is useful only if its stated uncertainty can be trusted [3, 34]. We express the standard uncertainty of the payload-compensated window-mean residual of joint i as a GUM-type budget,

$$u_i^2 = \underbrace{\frac{s_i^2}{n} + \max(\sigma_i^{\text{LOO}}, \sigma_i^{\text{ch}})^2}_{\text{Type A}} + \underbrace{\frac{a_i^2}{3} + (u_\kappa \bar{r}_i)^2}_{\text{Type B}}, \quad (8)$$

whose components are listed in Table 2. The Type A part is evaluated from Phase 1 data: s_i^2 is the pooled variance of the 0.1-s sub-window means within the calibration windows, n is the number of sub-windows averaged, and the model-error scale is the larger of the leave-one-out calibration residual σ_i^{LOO} and the challenge residual σ_i^{ch} , so that it reflects extrapolation to orientations not used for fitting. The Type B part covers effects that are invisible in contact-free calibration: residual dry friction after compensation, modeled as uniform with half-width a_i [15], and the tolerance u_κ of the torque constant, which scales the load-dependent part $\bar{r}_i$ of the residual.

The follower contact of (A3) does not reproduce the force exactly from one view to the next. We therefore model the force of view k as $\mathbf{f} + \delta_k$ with $\delta_k \sim \mathcal{N}(\mathbf{0}, \sigma_f^2 \mathbf{I}_3)$, $\sigma_f = \varepsilon_f \|\hat{\mathbf{f}}\|$, and marginalize δ_k in closed form in the likelihood of Section 3.4.

Table 3

Recovery rules. $\hat{N}_{\text{eff}} = 1 / \sum_i (w^{(i)})^2$; $\min_i S$ is the best profile residual; $\sigma_{\parallel}$ is the Cramér–Rao standard deviation along the current line of action. Every rule also draws $(1 - \eta)N$ particles from the current particle set q_0 and weights all draws by (10); the numbers of draws are given in parentheses ($\eta = 0.30$, $N = 3000$).

Rule	Failure mode	Trigger	Proposals q_j (draws n_j)
D1	Weight degeneracy	$\hat{N}_{\text{eff}}/N < \rho_{\text{th}}$, every update	Particle set, wider kernel (ηN)
D2	Narrow but inconsistent	$r_{95} < r_{\text{nar}}$ and $\min_i S > \tau_K$ for L consecutive updates	Line ($\eta N/3$), local ($\eta N/3$), uniform over $\mathcal{B}$ ($10\eta N/3$)
D3	Support loss along the line of action	$\sigma_{\parallel} \geq \ell_{\text{min}}$, end of each view	Line along $\hat{\mathbf{u}}_k = \mathbf{R}_k^T \hat{\mathbf{f}} / \ \hat{\mathbf{f}}\ $ (ηN)

3.4. Force-profiled contact particle filter

Each particle $\mathbf{p}^{(i)} \in \mathcal{B}$, $i = 1, \dots, N$, represents a contact-point hypothesis. The force is not sampled but eliminated by generalized least squares, with a single world-frame force shared by all views. For view k with window-mean residual $\bar{\mathbf{r}}_k$, weight $\mathbf{W}_k = \text{diag}(u_i^{-2})$, and $\mathbf{A}_k = \mathbf{J}_k^T$, define $\mathbf{T}_k(\mathbf{p}) = \mathbf{B}(\mathbf{p})^T \mathbf{A}_k^T \mathbf{W}_k \mathbf{A}_k \mathbf{B}(\mathbf{p})$ and $\mathbf{b}_k(\mathbf{p}) = \mathbf{B}(\mathbf{p})^T \mathbf{A}_k^T \mathbf{W}_k \bar{\mathbf{r}}_k$, where $\mathbf{B}(\mathbf{p}) = [\mathbf{I}_3; [\mathbf{p}]_{\times}]$. Marginalizing the follower variation by the Woodbury identity and the matrix determinant lemma gives the effective quantities $\tilde{\mathbf{T}}_k = \mathbf{T}_k - \mathbf{T}_k \mathbf{K}_k \mathbf{T}_k$, $\tilde{\mathbf{b}}_k = \mathbf{b}_k - \mathbf{T}_k \mathbf{K}_k \mathbf{b}_k$, $\mathbf{K}_k = (\sigma_f^{-2} \mathbf{I}_3 + \mathbf{T}_k)^{-1}$, and the profile likelihood

$$\hat{\mathbf{f}}(\mathbf{p}) = \left(\sum_k \mathbf{R}_k \tilde{\mathbf{T}}_k \mathbf{R}_k^T \right)^{-1} \sum_k \mathbf{R}_k \tilde{\mathbf{b}}_k, \quad S(\mathbf{p}) = \sum_k \left(\tilde{c}_k - \tilde{\mathbf{b}}_k^T \mathbf{R}_k^T \hat{\mathbf{f}} \right) + \log \det \left(\mathbf{I}_3 + \sigma_f^2 \mathbf{T}_k \right), \quad (9)$$

with $\tilde{c}_k = \bar{\mathbf{r}}_k^T \mathbf{W}_k \bar{\mathbf{r}}_k - \mathbf{b}_k^T \mathbf{K}_k \mathbf{b}_k$. Sharing $\mathbf{f}$ across views is not needed for identifiability (Remark 1), but under (A2) it pools the force information of all views. The follower-variation model interpolates between the two cases: $\sigma_f \rightarrow 0$ recovers a strictly shared force, and $\sigma_f \rightarrow \infty$ independent per-view forces. Weights follow $w^{(i)} \propto \exp(-\frac{1}{2} S(\mathbf{p}^{(i)}))$, which reduces a six-dimensional search to a three-dimensional one. The normalizer $\log \det(\mathbf{I}_3 + \sigma_f^2 \mathbf{T}_k)$ grows with the moment arm and therefore adds a weak preference for short moment arms along the line of action; this preference is a property of the noise model, not information from the data, which is why the recovery of Section 3.5 is triggered by the data information rather than by the shape of the posterior.

The residual is averaged over 0.1-s sub-windows, and each sub-window triggers one update, i.e., 15 updates per 1.5-s measurement window. Because the contact point is static, the filter is a resample–move sequential Monte Carlo sampler [27] for the sequence of posteriors $\pi_t(\mathbf{p}) \propto \exp(-\frac{1}{2} S_t(\mathbf{p}))$ on $\mathcal{B}$, where S_t is the profile objective (9) of the data up to sub-window t . Before each update, every particle takes one random-walk Metropolis–Hastings step with a Gaussian proposal of annealed width $\sigma_t = \max(\sigma_{\text{min}}, \sigma_0 0.9^t)$, $\sigma_0 = 5$ mm, $\sigma_{\text{min}} = 0.5$ mm, which is accepted with probability $\min\{1, \exp(-\frac{1}{2} [S_t(\mathbf{p}') - S_{t-1}(\mathbf{p})])\}$ and rejected outside $\mathcal{B}$. The step leaves π_{t-1} invariant; unlike kernel roughening [26], it therefore does not widen the target, and it rejects most steps that would move hypotheses off the narrow likelihood ridge around the line of action. The update then multiplies the weights by $\exp(-\frac{1}{2} [S_t(\mathbf{p}) - S_{t-1}(\mathbf{p})])$, and the particles are resampled systematically when the effective sample size falls below $N/2$. We use $N = 3000$ in a search box $\mathcal{B} = [-0.15, 0.15] \times [-0.15, 0.15] \times [-0.05, 0.30]$ m in the tool frame. The posterior mean $\hat{\mathbf{p}}$, covariance $\Sigma_{\mathbf{p}}$, and the radius r_{95} of the ball around $\hat{\mathbf{p}}$ containing 95% of the weight summarize the measurement and its uncertainty.

3.5. Ambiguity-preserving recovery

Once a hypothesis disappears from a finite particle set, no later measurement can restore it. Three rules monitor different ways in which support is lost (Table 3), and all three re-supply hypotheses through the same importance-weighted step.

Importance-weighted re-supply Re-supply should change where the particles are, not the distribution that they target. Each rule therefore replaces the particle set by an importance sample from the defensive mixture

$$q(\mathbf{p}) = \frac{1}{M} \left(n_0 q_0(\mathbf{p}) + \sum_j n_j q_j(\mathbf{p}) \right), \quad M = n_0 + \sum_j n_j, \quad (10)$$

where q_0 is the current weighted particle set smoothed by a Gaussian kernel of width σ_t , drawn $n_0 = (1 - \eta)N$ times, and q_j are the proposals of the rule, drawn n_j times (Table 3); η is the re-supply fraction. Each of the M draws $\mathbf{p}^{(m)}$

receives the importance weight $w^{(m)} \propto \pi_i(\mathbf{p}^{(m)})/q(\mathbf{p}^{(m)})$, and the weighted set is resampled to N particles. Because the weights are the ratio of target to proposal, the weighted set is a consistent importance-sampling approximation of π_i whatever the proposal: a region that the cloud had lost receives weight in proportion to its posterior probability, and re-supply does not systematically sharpen the target without new data. For a finite particle set, the approximation carries Monte Carlo error, which a no-data test quantifies (Appendix A): applied ten times in succession to filter states after one and two views, every rule left r_{95} at a median of 1.00 and 0.99 times that of an independent reference sample of the same posterior.

D1: weight degeneracy When the effective sample size collapses, D1 draws ηN particles from the current particle set smoothed with a wider kernel of $\max(3\sigma_i, 2 \text{ mm})$. D1 restores sample diversity around the current mode but cannot recover a truth that already lies outside the particle cloud.

D2: narrow but inconsistent A posterior can be narrow and have a healthy effective sample size while no particle explains the measurements; this is the path to a CBW measurement, and D1 does not detect it. D2 triggers when a narrow posterior ($r_{95} < r_{\text{nar}}$) remains inconsistent with the data for L consecutive updates, so that transient outliers do not reset the filter. Inconsistency means that the best particle score exceeds the threshold τ_K , the 99% quantile of the χ^2 distribution with $\max(1, 7K - 6)$ degrees of freedom after K views. Because $S(\mathbf{p})$ contains the normalizer of Eq. (9) and the minimum is taken over a finite particle set, τ_K is not a nominal significance level; it is a chi-square-scaled inconsistency threshold that was frozen on the development set, and its false-alarm behavior is part of the evaluated method. D2 uses three proposals: a line proposal along the chord of $\mathcal{B}$ through $\hat{\mathbf{p}}$ in the direction of the current line of action (the geometrically meaningful escape direction), uniform along the chord and Gaussian perpendicular to it with the current perpendicular posterior spread; the current particle set smoothed with a 10-mm kernel; and a uniform proposal over $\mathcal{B}$. The line and local proposals receive $\eta N/3$ draws each, and the uniform proposal $10\eta N/3$ draws, so that the global component covers the whole search region densely; with $\eta = 0.30$ and $N = 3000$, D2 draws 2100 particles from q_0 , 300 each from the line and local proposals, and 3000 uniformly, and the mixture (10) weights the components by these counts.

D3: line-of-action support preservation Between views, the posterior is correctly elongated along the unresolved direction (Proposition 1), and resampling thins it along exactly that direction. D3 is evaluated once at the end of each view and triggers on the information content of the data, not on the shape of the particle cloud: it computes the Cramér–Rao standard deviation $\sigma_{\parallel} = (\hat{\mathbf{u}}_k^T \mathbf{I}_{p|f}^{-1} \hat{\mathbf{u}}_k)^{1/2}$ along the current line of action from the profile information of the views collected so far (Section 3.6), and fires when $\sigma_{\parallel} \geq \ell_{\min}$, i.e., when the data cannot yet localize the contact along the line. After a single view, $\sigma_{\parallel}$ is unbounded by Proposition 1, so D3 always fires, even if the particle cloud has already collapsed onto a short stretch of the line. It draws ηN particles from the line proposal along $\hat{\mathbf{u}}_k$: samples $\hat{\mathbf{p}} + s \hat{\mathbf{u}}_k + \mathbf{v}$, where s is uniform over the chord of $\mathcal{B}$ through $\hat{\mathbf{p}}$ along $\hat{\mathbf{u}}_k$ and $\mathbf{v}$ is a Gaussian perturbation perpendicular to $\hat{\mathbf{u}}_k$ with the current perpendicular posterior spread. The direction $\hat{\mathbf{u}}_k = \mathbf{R}_k^T \hat{\mathbf{f}} / \|\hat{\mathbf{f}}\|$ is the theoretical line of action computed from the profiled force, not the empirical principal axis of the cloud, which is biased by the very thinning that D3 counteracts. Because the re-supplied particles lie in the null direction of the current view, the data of that view do not discriminate among them, apart from the weak normalizer preference of Section 3.4, and their importance weights are nearly uniform along the line; only a new view can resolve it. D3 therefore does not resolve the ambiguity but keeps the correct hypothesis alive until the active view resolves it.

Execution order and guards Within each update, the filter moves the particles by one Metropolis–Hastings step, profiles the force, updates the weights and diagnostics, evaluates D2 and then D1, and otherwise resamples when $\hat{N}_{\text{eff}} < N/2$; D3 is evaluated at the end of each view. A cooldown of five updates after each activation and caps of two activations per view and eight per episode prevent excessive recovery. Table 4 lists all parameters, which were frozen on a separate development set of 200 trials before the evaluation reported below.

3.6. Active measurement configuration

Profile information Linearizing the measurement model (3) in $\xi = (\mathbf{p}, \mathbf{f})$ gives the Jacobian

$$\mathbf{H}_k(\mathbf{p}, \mathbf{f}) = \begin{bmatrix} \mathbf{J}_k^T \begin{bmatrix} \mathbf{0}_3 \\ -[\mathbf{f}_k]_{\times} \end{bmatrix} & \mathbf{G}_k(\mathbf{p}) \end{bmatrix} \in \mathbb{R}^{7 \times 6}, \quad \mathbf{I}^{(\mathcal{K})} = \sum_{k \in \mathcal{K}} \mathbf{H}_k^T \tilde{\mathbf{W}}_k \mathbf{H}_k = \begin{bmatrix} \mathbf{I}_{pp} & \mathbf{I}_{pf} \\ \mathbf{I}_{fp} & \mathbf{I}_{ff} \end{bmatrix}, \quad (11)$$

Table 4

Frozen estimator parameters.

Parameter	Value	Parameter	Value
Particles N	3000	D2 persistence L	10 updates
D1 ratio ρ_{th}	0.02	D3 threshold ℓ_{min}	0.03 m
Re-supply fraction η	0.30	Min. MH step σ_{min}	0.5 mm
D2 radius r_{nar}	0.05 m	Follower variation ϵ_f	3%
Observer gain K_O	50 s ⁻¹	Gate scale σ_g	0.1 N m

where $\tilde{W}_k$ is the view weight including the marginalized follower variation. The weight is held fixed at the current estimate, so (11) is the fixed-covariance (Gauss–Newton) approximation of the Fisher information; it neglects the weak dependence of the marginalized covariance on $\mathbf{p}$ and $\mathbf{f}$, which enters only through the follower-variation term. Since the force is unknown but the quantity of interest is the contact point, we evaluate the profile (Schur-complement) information $\mathbf{I}_{p|f} = \mathbf{I}_{pp} - \mathbf{I}_{pf} \mathbf{I}_{ff}^{-1} \mathbf{I}_{fp}$, which discounts the part of the data that a change of force could explain [33].

Posterior-expected score Evaluating (11) at the posterior mean alone ignores that the posterior is a line whose points yield different information. For a candidate orientation $\mathbf{R}^c$, we therefore score the information expected over the posterior,

$$\mathcal{J}(\mathbf{R}^c) = \frac{1}{M} \sum_{i=1}^M \log \det \left(\mathbf{I}_{p|f}^{(1:k) \cup \{c\}} (\bar{\mathbf{p}}^{(i)}, \hat{\mathbf{f}}(\bar{\mathbf{p}}^{(i)})) + \epsilon \mathbf{I}_3 \right) - \lambda_m \varphi(\mathbf{R}_k, \mathbf{R}^c), \quad (12)$$

where $\{\bar{\mathbf{p}}^{(i)}\}$ is a systematic subsample of size $M = 64$ of the posterior, $\epsilon = 1 \text{ m}^{-2}$, φ is the rotation angle of the transition, and $\lambda_m = 1 \text{ rad}^{-1}$ penalizes large motions that would endanger the follower contact. The score evaluates the cumulative information of past views and the candidate jointly, so a candidate that repeats the current line of action is not rewarded.

Candidates inside the validated envelope The payload compensation is validated only up to the challenge tilt of Phase 1. Candidates are therefore generated relative to the initial orientation, not the current one: tilts of 10°, 20°, and 30° about 26 axes (the directions of a $3 \times 3 \times 3$ lattice). Candidates within 5° of an already visited view are dropped, and those that violate joint limits or would require more than 85% of a joint torque limit under the estimated payload and contact wrench are discarded.

Position-preserving reorientation The end effector moves to the selected orientation along eight inverse-kinematics waypoints on the rotation geodesic, interpolated by a cubic spline with a 2.0-s minimum-jerk time scaling, while its position is held at $\mathbf{o}_k$. Keeping the position fixed changes the object-frame force direction, and hence the moment-arm geometry, without dragging the object and its contact across the workspace, which preserves (A1) and (A3). The rotation can equally be performed about the current estimate $\hat{\mathbf{p}}$ instead of the tool center point, which shortens the path of the contact; Section 4.7 compares both with physical contact. Null-space motion alone is ineffective: if the end-effector pose is unchanged, so is the measurement geometry. Localization updates are suspended during the transition and resume after a settling time of 1.0 s, over a window of 1.5 s, consistent with (A4). An episode ends when at least two views have been taken and r_{95} falls below 10 mm without D3 firing, or after five views.

4. Experimental evaluation

4.1. Protocol

Simulation setup We simulated a Franka Research 3 in MuJoCo 3.13 at 1 kHz, with the link inertial parameters of the MuJoCo Menagerie FR3 model and a 0.73-kg Franka Hand, which constitute the nominal model used by the controller and the estimator. The simulated plant deviates from the nominal model in the ways that a commanded-torque-only arm does (Table 5). The arm is torque-controlled by a joint PD law with nominal gravity and Coriolis compensation; the observer runs on the nominal model with measured signals. Each trial draws a random initial configuration around the Franka ready pose, a rigid box-shaped object of $0.04\text{--}0.10 \times 0.04\text{--}0.10 \times 0.10\text{--}0.22$ m with a lateral offset of up to

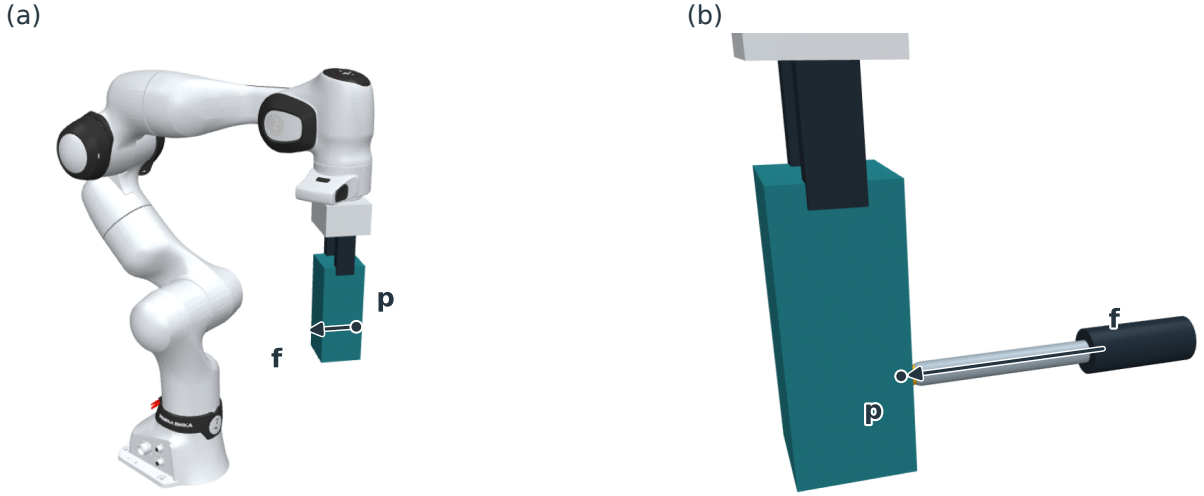


Figure 3: Simulation apparatus, rendered in MuJoCo. (a) The FR3 holds an unknown object, with a force $\mathbf{f}$ applied at a fixed material point $\mathbf{p}$. The main study varies the object mass (0.3–1.5 kg), contact point, force (5–25 N), and model errors; only joint states and commanded torques enter the estimator. (b) In the physical-contact extension, the force arises from contact with a spherical tip of radius 6 mm, whose lateral compliance is varied (Section 4.7). Between measurement windows the robot reorients the object; each 2.0-s move is followed by 1.0 s of settling and a 1.5-s measurement window, with at most five views. The panels illustrate the apparatus.

Table 5

Plant deviations from the nominal model (per trial). $\mathcal{N}$: normal, $\mathcal{U}$: uniform.

Error source	Distribution
Link masses	$\times(1 + \mathcal{N}(0, 0.03^2))$, clipped to $\pm 7.5\%$
Link centers of mass	$+\mathcal{N}(0, (3 \text{ mm})^2)$ per axis
Joint viscous damping	$\times(1 + \mathcal{U}(-0.2, 0.2))$
Residual dry friction	$\mathcal{U}(0.02, 0.06) / \mathcal{U}(0.01, 0.03) \text{ N m (joints 1–4 / 5–7)}$
Torque constant	$\times(1 + \mathcal{N}(0, 0.01^2))$
Torque tracking noise	white, $0.2 / 0.05 \text{ N m (joints 1–4 / 5–7)}$
Encoder / velocity noise	$2 \times 10^{-5} \text{ rad} / 2 \times 10^{-3} \text{ rad s}^{-1}$
Follower force	2% magnitude ripple, 1° direction wobble

30 mm and a mass of 0.3–1.5 kg, a contact point uniformly on its free surface, and a force of 5–25 N whose direction lies within 60° of the inward surface normal. Fig. 3 illustrates the simulated apparatus and the physical-contact extension. Conditions that would require more than 80% of a joint torque limit at the initial orientation or at 30° tilts are redrawn. The contact force is applied at the material point and follows it during reorientation, with a 2% magnitude ripple and a 1° direction wobble (Ornstein–Uhlenbeck processes with a 0.5-s time constant) that emulate an imperfect follower. The Monte Carlo study comprises 1,200 paired trials per variant on test seeds that are disjoint from the 200 development trials used to freeze the design.

Metrics Position accuracy is reported as the median, mean, and 95th-percentile Euclidean error $e_p = \|\hat{\mathbf{p}} - \mathbf{p}\|$, and as the success rate $e_p < 10 \text{ mm}$. Force accuracy is reported as the relative magnitude error and the angular error of $\hat{\mathbf{f}}$. The consistency of the reported uncertainty is assessed by the coverage of the 95% credible ellipsoid, i.e., the fraction of trials with $(\hat{\mathbf{p}} - \mathbf{p})^\top \Sigma_p^{-1} (\hat{\mathbf{p}} - \mathbf{p}) \leq \chi_{0.95}^2(3)$, a NEES-type test [35], and by the CBW rate. The ellipsoid is the Gaussian region of the posterior covariance; the radius r_{95} of the weighted-particle ball, which the stopping rule and Definition 1 use, describes a different region, and the two are reported separately.

Definition 1 (Confident-but-wrong measurement). An episode is CBW if its final posterior is confident, $r_{95} < r_{\text{conf}}$, while its estimate is wrong, $e_p > e_{\text{tol}}$. We use $r_{\text{conf}} = 10 \text{ mm}$ and $e_{\text{tol}} = 20 \text{ mm}$.

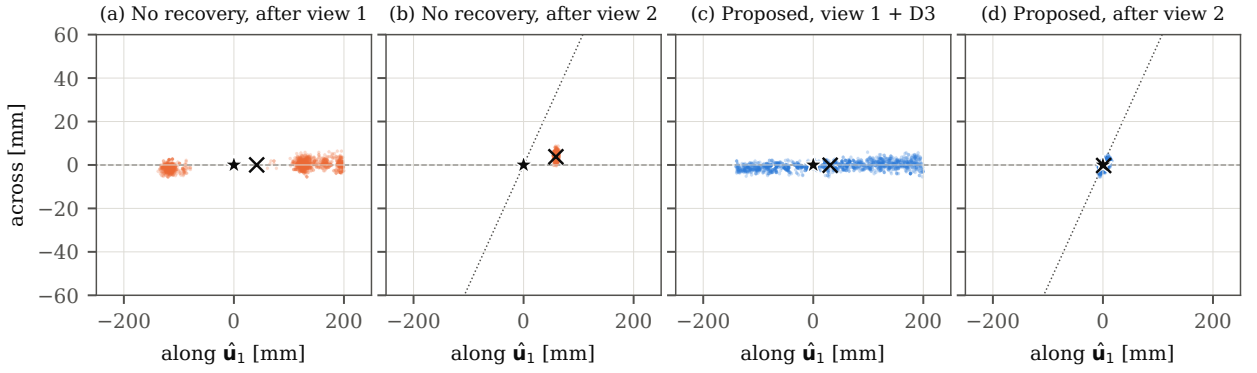


Figure 4: Representative paired episode (development seed), projected on the plane spanned by the first-view line of action (dashed) and the second-view line of action (dotted); star: true contact, cross: posterior mean. (a,b) Without recovery, the particles that survive the first view cover two stretches of the line that miss the truth; after the second view, the posterior is narrow and wrong (CBW, 59.8 mm error). (c,d) With recovery, D3 re-supplies hypotheses along the whole line after the first view, and the second view selects the true contact (1.3 mm error). The two axes are scaled differently.

Confidence and accuracy are thus evaluated separately: a CBW measurement is more harmful than a wide posterior, because downstream use has no indication that it should not be trusted. Efficiency is reported as the number of views, the time from contact onset to the end of localization, and the computation time of the next-view selection. Confidence intervals (95%) are obtained by bootstrap over trials.

Baselines and ablations All variants share the observer, the payload calibration, the uncertainty budget, and the frozen parameters of Table 4. (i) The classical regularized particle filter, which roughens the particles with the Gaussian kernel of Section 3.4 [26] instead of the Metropolis–Hastings move, without recovery; (ii) CPF without recovery; (iii) CPF with a single recovery rule D1, D2, or D3; (iv) D3 re-supplying along the empirical principal axis ν_1 of Σ_p , triggered by cloud shape ($\lambda_1 \geq 3\lambda_2$ and $\sqrt{\lambda_1} \geq \ell_{\min}$), as a shape-based alternative to the information-based D3; (v) next view chosen at random among feasible candidates, by maximal rotation, by FIM at the posterior mean, or by the posterior-expected score (12); (vi) the proposed method (D1+D2+D3 with the posterior-expected score). Paired trials share the robot, object, contact point, force, model and actuation errors, and plant noise sequence, and every variant branches from the same post-calibration state. Unless stated otherwise, the results evaluate the estimator on all trials of a study, without the trust gate; Section 4.6 reports the complete pipeline including the gate.

4.2. Observability

Single view Single-view posteriors aligned with the theoretical line of action, as predicted by Proposition 1. The principal axis of Σ_p deviated from the true object-frame force direction by a median of 4.8° , with a median axis ratio $\sqrt{\lambda_1/\lambda_2}$ of 6. Perpendicular to the line, the single-view error had a median of 11.5 mm, whereas along the line it was bounded only by the extent of $\mathcal{B}$. Fig. 4 shows a representative episode.

Two views Table 6 and Fig. 2 report two views whose second orientation is rotated by ϑ about an axis perpendicular to the true force, with recovery enabled and no early stopping. The error is decomposed along the worst-information direction of Corollary 1, i.e., the bisector of the two lines of action, and in the plane orthogonal to it. In a noise-limited control study without model or actuation errors and with five times the torque noise, so that noise dominates the particle resolution, the RMS error along the worst direction followed the $1/\sin(\vartheta/2)$ scaling of (5) from 45° down to 10° (normalized values 1.47, 2.17, and 3.92 at 30° , 20° , and 10° , against 1.48, 2.20, and 4.39) and fell below it at 5° (6.63 against 8.78), where the error along the nearly parallel lines begins to be limited by the search region. Under the realistic errors of Table 5, the same component still grew steeply as the two lines approached each other, from 10.4 mm at 45° to 30.8 mm at 5° , whereas the orthogonal error stayed between 7.9 and 10.2 mm at every angle. This floor is the bias of the model and actuation errors, which the noise-based bound does not contain and which no choice of view can remove; it is why the total error saturates at moderate angles and why the candidate tilts need not exceed 30° . At $\vartheta = 0$, the second view repeats the first, and the error along the line is bounded only by the search region.

Table 6

Two-view error versus the inter-view force angle θ (rotation axis perpendicular to $\mathbf{f}$). Worst direction: bisector of the two lines of action (Corollary 1). Normalized values are divided by their value at $\theta = 45^\circ$.

θ	0°	5°	10°	20°	30°	45°
Bound (5), normalized	∞	8.78	4.39	2.20	1.48	1.00
Noise-limited control study (300 trials)						
RMS error, worst direction, normalized	—	6.63	3.92	2.17	1.47	1.00
Realistic errors (300 trials)						
RMS error, worst direction [mm]	56.6	30.8	28.4	14.9	11.5	10.4
RMS error, orthogonal plane [mm]	7.9	9.0	10.2	8.9	9.4	9.9
Median total error [mm]	33.0	16.5	13.2	10.1	9.4	8.8

4.3. Recovery and measurement reliability

Table 7 isolates the particle move and the recovery rules, and Fig. 6 shows the error distributions. Relative to the classical regularized particle filter, the proposed method reduced the p95 error from ?? mm to 21.9 mm (paired bootstrap 95% CI of the difference: ?? mm) and the CBW rate from ??? to 0.8% (10 of 1,200 trials; exact 95% CI 0.4–1.5%). Both layers of support preservation contribute. The Metropolis–Hastings move alone lowered the CBW rate to 17.7%: kernel roughening pushes hypotheses off the narrow likelihood ridge around the line of action, where the next update removes them, whereas the Metropolis–Hastings move rejects most such steps and keeps the hypotheses on the ridge.

The independent contribution of recovery is measured against this stronger filter, on the same trials. Recovery lowered the CBW rate from ??? to ??? (?? trials CBW only without recovery, ?? only with it; exact McNemar p ??), the rate of errors above 20 mm from ??? to ??? (paired 95% CI of the reduction: ?? percentage points), and the p95 error from 137.6 mm to 21.9 mm (CI of the difference: 102.1–126.7 mm), while the median changed from 9.2 mm to 6.6 mm: recovery acts on the tail, where the measurement is most harmful. It also made the reported uncertainty trustworthy: the 95% credible ellipsoid contained the true contact in ??? of the trials, against ??? without recovery and ??? for the regularized filter, and the 95% particle ball of radius r_{95} in ???, against ??? (paired CI of the gain: ?? percentage points). The reduction of the CBW rate does not hinge on the thresholds of Definition 1: with $(r_{\text{conf}}, e_{\text{tol}}) = (10, 10)$, (15, 30), and (20, 40) mm, the CBW rate was 23.0, 15.8, and 13.9% without recovery and 6.2, 0.4, and 0.2% with the proposed method.

The classical effective-sample-size rule D1 alone made the tail worse (p95 170.3 mm, CBW 28.0%). The dominant failures are therefore not a collapse of the weights, but the loss of support along the unresolved line of action and narrow posteriors that no longer explain the data. The force dependence of the failures confirms this mechanism. Without recovery, the CBW rate grew with the force magnitude, from 0.7% at 5–10 N to 40.7% at 20–25 N (38.8% in the upper third of force magnitudes): the larger the force, the sharper the likelihood perpendicular to the line, and the fewer particles survive on it. With recovery, the CBW rate stayed at or below 2.5% in every force bin (1.8% in the upper third).

The two targeted rules address the two failures on different time scales. D2 is a safety net that acts once a narrow posterior has persistently contradicted the data; alone, it reduced the final p95 error to 22.3 mm with a CBW rate of 1.2%. D3 acts at the end of the first view, before any wrong commitment, and makes the second view count for the bulk of the trials: after two views, 61.2% of the trials were below 10 mm with D3 alone, against 44.8% with D2 alone and 63.1% with both. The tail after two views is set by the trials that D2 repairs later, with p95 errors of 35.5 mm for D3 alone, 33.0 mm for D2 alone, and 25.9 mm for both. Combining the rules gave the lowest final CBW rate and p95 error of all variants. How D3 is triggered and where it re-supplies matter. The shape-based variant, which re-supplies along the empirical principal axis only while the cloud is still elongated, barely improved the tail (p95 132.6 mm, CBW 12.2%): by the end of the first view, the surviving cloud has often already collapsed onto a short stretch of the line, is no longer elongated, and the shape test does not fire.

Fig. 5 extends the coverage test to all credible levels. With recovery, the empirical coverage follows the nominal level closely up to intermediate levels but is optimistic in the tail: the nominal 95% ellipsoid contained the true contact in 81.9% of the trials, so that the miss rate of 18.1% was 3.6 times the nominal 5%; without recovery, the reported uncertainty is optimistic at every level, because a lost hypothesis cannot be represented by any spread of the surviving particles.

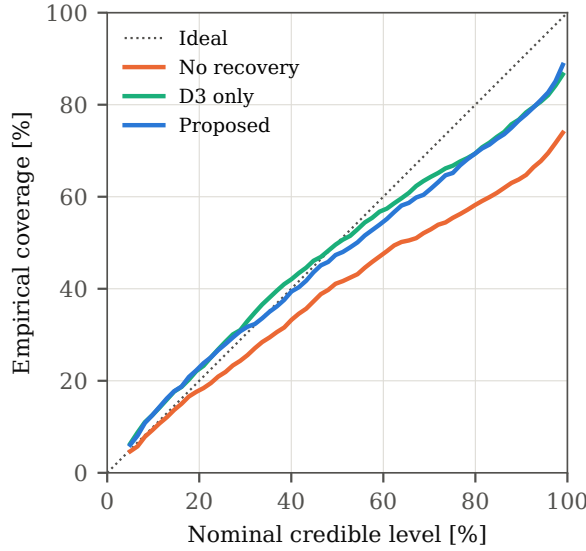


Figure 5: Reliability of the reported uncertainty: empirical coverage of the credible ellipsoid versus its nominal level (NEES test, 1,200 trials).

Table 7

Recovery ablation (simulation, posterior-expected next-view selection in all rows, 1,200 paired trials). The first row replaces the Metropolis–Hastings move by kernel roughening. Cov.: fraction of trials in which the 95% credible ellipsoid, or the 95% particle ball of radius r_{95} , contains the true contact.

Recovery	Median [mm]	p95 [mm]	$e_p > 20$ mm [%]	CBW [%]	Cov. ellipsoid [%]	Cov. r_{95} [%]	Views
None, regularized PF	??	??	??	??	??	??	??
None	9.2	137.6	??	17.7	68.5	??	4.27
D1	11.2	170.3	??	28.0	61.0	??	4.10
D2	7.2	22.3	??	1.2	77.7	??	4.22
D3, shape-based (ν_1)	7.2	132.6	??	12.2	79.6	??	4.41
D3, information-based ($\hat{\mu}_k$)	7.0	31.9	??	6.6	81.2	??	4.42
D1+D2+D3 (proposed)	6.6	21.9	??	0.8	81.9	??	4.21

The stopping rule inherits this reliability. With the proposed method, ??% of the episodes stopped before the fifth view; at the stop, the true contact lay inside the reported radius r_{95} in ??% of them, against the nominal 95%, and ?? of the ?? stops had an error above 20 mm (exact 95% upper bound ??%). Without recovery, the reported radius contained the true contact at only ??% of the stops, and with the regularized particle filter at ??%: their stops are declared on posteriors that have lost the truth.

4.4. Active sensing

Table 8 crosses recovery with the next-view strategy, and Fig. 7 shows the error after each view. Without recovery, neither strategy removes the tail, because the true hypothesis is often lost before the second view is taken. With recovery, both strategies eventually reach similar final errors under the stopping rule, but the active strategy stops earlier (4.21 versus 4.64 views on average; 17.0 s versus 18.9 s from contact onset). At a fixed budget, the difference is large: after two views, the p95 error was 25.9 mm with the active view and 40.4 mm with a random view (paired difference 14.4 mm, 95% CI 10.9–20.5 mm), and 63.1% versus 45.6% of the trials were below 10 mm. Recovery preserves the right hypothesis; active sensing separates it with the fewest measurements. The stopping threshold trades accuracy for speed: with $r_{95} < 20$ mm instead of 10 mm, localization took 2.83 views and 10.7 s instead of 4.21 views and 17.0 s, at a median error of 7.3 mm, a p95 error of 22.3 mm, and a CBW rate of 2.8% for the correspondingly relaxed confidence radius of 20 mm.

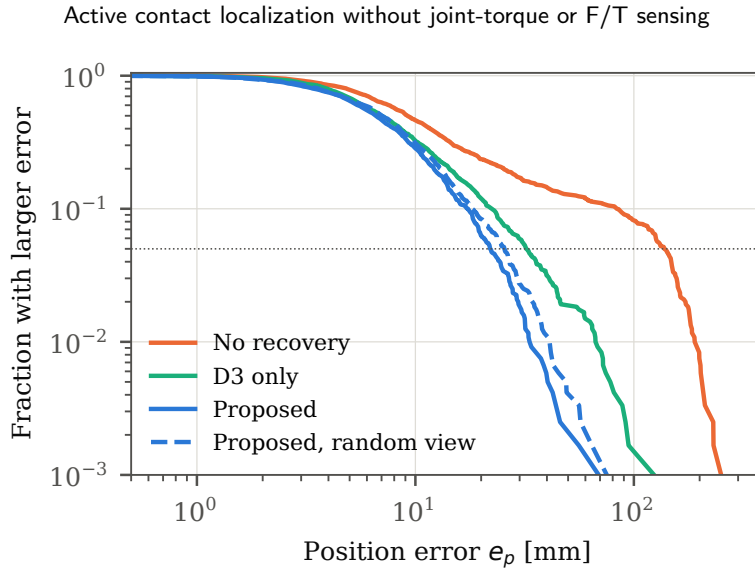


Figure 6: Complementary cumulative distribution of the position error over 1,200 paired trials (log-log). The dotted line marks the 95th percentile.

Table 8

Recovery $\times$ next-view strategy (simulation, 1,200 paired trials per cell).

Recovery	Next view	p95 [mm]	CBW [%]	$e_p < 10$ mm [%]	Views
Off, regularized PF	Random	??	??	??	??
Off, regularized PF	Active	??	??	??	??
Off	Random	154.8	17.5	44.7	4.51
Off	Active	137.6	17.7	53.8	4.27
On	Random	25.5	1.7	68.9	4.64
On	Active	21.9	0.8	71.7	4.21

Table 9

Next-view selection rules (simulation, recovery on, 1,200 paired trials). Selection: mean wall time per selection, including the inverse kinematics of all candidates, measured during the parallel Monte Carlo runs; single-core times are given in the text.

Next view	Median [mm]	p95 [mm]	CBW [%]	Views	Selection [s]
Random feasible	7.0	25.5	1.7	4.64	1.12
Maximal rotation	7.1	23.0	1.6	4.49	1.12
FIM at posterior mean	6.6	22.8	1.1	4.20	1.18
Posterior-expected FIM (12)	6.6	21.9	0.8	4.21	1.58

Among the view-selection rules (Table 9), maximal rotation performed like random selection: large rotations about axes nearly parallel to the force add motion without adding information (Proposition 2). The posterior-expected score performed on par with the FIM evaluated at the posterior mean (after two views, p95 25.9 mm versus 25.2 mm), so the cheaper mean-point score is an equally valid choice in this regime. On a single idle CPU core, generating and checking the 78 feasible candidates by inverse kinematics took 303 ms, and scoring them took 99 ms with the posterior-expected score and 10 ms with the mean-point score. Because the candidate set is fixed relative to the initial orientation, the inverse kinematics can be computed once per episode, which leaves the scoring well inside the 1-s settling period that follows each reorientation.

4.5. Robustness, sensitivity, and computation

The accuracy scaled with the force magnitude as predicted by the $1/\|f\|$ factor of (5) (Table 10): the median error fell from 9.8 mm at 5–10 N to 5.0 mm at 20–25 N. The force itself was measured with a median magnitude error of

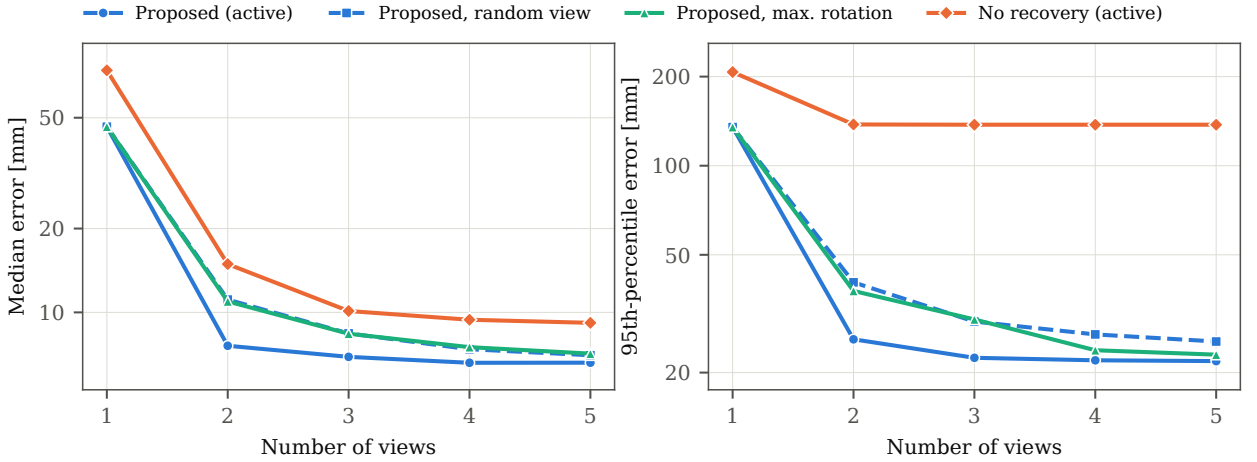


Figure 7: Median and 95th-percentile position error after each view (1,200 paired trials; trials that stopped earlier keep their final estimate). With recovery, the actively selected second view already achieves the accuracy that random or maximal rotations reach only after four to five views.

Table 10

Stress conditions (simulation, proposed method). Force errors are medians. Force bins and the heavy-payload row are subsets of the main study; the other rows are separate studies. In the last two rows, the true error exceeds the value assumed by the uncertainty budget.

Condition	Trials	Median [mm]	p95 [mm]	CBW [%]	Cov. [%]	$ \Delta\ f\ $ [%]	$\angle\Delta f$ [deg]
$\ f\ \in [5, 10)$ N	288	9.8	30.3	0.3	85.4	2.0	1.6
$\ f\ \in [10, 15)$ N	324	7.0	18.5	0.0	86.1	1.4	1.1
$\ f\ \in [15, 20)$ N	283	5.7	18.1	2.5	79.5	1.1	0.9
$\ f\ \in [20, 25)$ N	305	5.0	13.9	0.7	76.4	1.0	0.8
Payload ≥ 1.1 kg	418	6.5	23.7	0.5	82.8	1.5	1.1
Link-mass error 10%	300	9.7	28.4	2.3	80.0	2.0	1.7
Residual friction $\times 3$	300	6.5	21.0	0.0	82.3	1.2	0.9
Torque constant 3% (budget 1%)	300	11.4	36.9	1.7	57.7	2.4	1.7
Follower 5%, 3° (budget 3%)	300	7.1	23.9	1.0	68.3	1.9	1.8

1.3% and a median angular error of 1.0°. Heavy payloads did not degrade the measurement, and neither did a threefold residual friction (median 6.5 mm, CBW 0.0%). A link-mass error of 10%, which the four-parameter payload calibration can absorb only locally, raised the median error to 9.7 mm and the CBW rate to 2.3%. When the true errors exceeded the values assumed in the uncertainty budget, the accuracy degraded gracefully but the reported uncertainty became optimistic. A torque-constant error of 3% (budget 1%) raised the median error to 11.4 mm and lowered the coverage from 81.9% to 57.7%, and a follower variation of 5% and 3° (budget 3%) lowered the coverage to 68.3% at a median error of 7.1 mm. The CBW rate stayed at or below 1.7% in both cases, because recovery does not rely on an exact budget.

Recovery is insensitive to most of its frozen parameters. In a separate study of 300 trials that changed one parameter at a time, the re-supply fraction ($\eta = 0.1$ and 0.5), the D3 threshold ($\ell_{\min} = 1$ and 10 cm), and a shorter D2 persistence ($L = 5$) kept the p95 error between 19.5 and 22.8 mm and the CBW rate at or below 1.0%. A longer persistence ($L = 20$) raised the CBW rate to 6.3%: it exceeds the 15 updates of one measurement window, so that D2 can no longer act within the view in which an inconsistency arises. The persistence must therefore stay below the number of updates per window.

With 500, 1000, 3000, and 5000 particles (400 matched trials), the p95 error was 22.1, 19.9, 19.2, and 18.9 mm and the CBW rate 2.2, 1.8, 0.2, and 1.0%. One filter update took 4.1, 7.7, 22.4, and 37.9 ms on a single CPU core in our Python implementation, so that $N = 3000$ uses less than a quarter of the 100-ms update period.

4.6. Trust gate

The gate study comprised 1,000 trials whose model errors were drawn from a mixture of link-mass errors of 2–15%, center-of-mass errors of 2–10 mm, torque-constant errors of 0–8%, and unmodeled joint springs of 0–3 N m/rad. A trial was labeled invalid when the RMS static compensation error over 40 random orientations inside the validated envelope exceeded 0.25 N m, which applied to 90.4% of the trials. One half of the trials derived the threshold by the lexicographic rule of Section 3.3. On the other half, the frozen gate ($\gamma = 0.87$) admitted 0 of the 456 invalid models (exact 95% upper bound 0.8%) and 0 CBW localizations, and it accepted 52.3% of the 44 valid models; the statistic separated valid from invalid models with an AUC of 0.99. The gate is deliberately conservative: rejecting a valid model delays the measurement, whereas admitting an invalid one produces a wrong measurement without warning. Applied to the main study, the frozen gate admitted 558 of the 1,200 trials (46.5%). On the admitted trials, the complete pipeline attained a median error of 5.2 mm, a p95 error of 15.8 mm, a CBW rate of 0.4%, and an ellipsoid coverage of 84.6%, against a CBW rate of 24.9% without recovery on the same trials. The gate screens exactly the trials that would yield the least reliable measurements: over the whole study, the median localization error was 7.1 mm for valid and 17.3 mm for invalid models, with CBW rates of 0.0% and 2.1%.

4.7. Physical contact and pusher compliance

The studies above apply the contact force as an ideal follower force at the material point. To test this abstraction, 300 additional trials replaced it by a physical contact: a pusher with a 6-mm spherical tip and a friction coefficient of 1.0 pressed on the object surface through the contact model of MuJoCo, and the contact force followed from the collision. The pusher applied the commanded world force either as a pure force controller that follows the surface (follower) or with a lateral impedance of 50, 150, or 300 N/m around its initial position (anchored), which models a pusher that does not follow the moving surface. The initial force direction was drawn within 15° of the inward normal, and the object was rotated about the tool center point as before, about the current estimate $\hat{p}$, or, as an ideal reference, about the true contact point (Table 11).

With a following pusher, the physical contact reproduced the results of the ideal force model: the median error was 7.8 mm, the p95 error 20.4 mm, and the CBW rate 0.7%. The force changed by a median of 3.6% between windows, and the tip slid by a median of 3.1 mm, because the constant world force approaches the edge of the friction cone as the surface rotates; in 0.7% of the trials, the accumulated rotation took the force outside the cone, and the tip slid off the face. Rotating about the estimate instead of the tool center point shortened the distance that the pusher had to follow from 48 to 24 mm at unchanged accuracy.

A pusher that does not follow turns the motion of the material point into a lateral force. The force then differed between views by a median of 9.7, 26.7, and 48.5% for lateral stiffnesses of 50, 150, and 300 N/m, which violates assumption (A2) and raised the CBW rate to 8.3, 19.7, and 38.0%. Rotating about the true contact point removed the problem (CBW 1.7%), but rotating about the estimate did not: after one view, the estimate is uncertain along the line of action by construction (Proposition 1), so an informative rotation about it still moves the true contact, and with a wrong pivot, the pusher lost contact in 27.3% of the trials. Contact-preserving measurement therefore requires a pusher that is compliant tangentially to the surface, such as a human hand or a force-controlled pusher without lateral position feedback; this is the practical meaning of assumption (A3). The reference protocol of the next section removes the contact mechanics altogether and thereby isolates the measurement chain.

4.8. Added-mass reference protocol

The weight of a rigidly mounted mass acts on the object as an equivalent point force at its center of mass, whose point of application and force are known by construction: the point is fixed in the object, the force has a constant world-frame direction, and no contact mechanics is involved. Both references are traceable, the force to a weighed mass and the point to the machined mounting geometry. The protocol thus localizes the point of application of an equivalent point force rather than a contact surface between parts, and it serves as a reference for the measurement chain. The estimator is not told that the force is a weight; it estimates the direction and magnitude of the force together with its point of application, exactly as for a contact. The protocol satisfies (A1)–(A3) exactly, so that the reference measures the complete sensorless measurement chain: the observer residual, the payload compensation, the uncertainty budget, and the estimator. We use this apparatus for the real-robot validation (Section 4.9) and simulated the complete protocol beforehand.

In simulation, a steel disc with a radius of 60 mm and a mass of 0.5, 1.0, or 1.5 kg was mounted, tilted by 45° , at one of eight positions of an oblique arm on a $40 \times 40 \times 120$ -mm bracket of 0.35 kg held in the gripper; hand, bracket,

Table 11

Physical contact (simulation, proposed estimator, 300 paired trials per row). Pivot: point about which the object is rotated. Travel: median of the largest displacement of the pusher tip during a reorientation. Force change: median of the largest change of the window-mean contact force relative to the first window.

Pusher	Pivot	Median [mm]	p95 [mm]	CBW [%]	Contact lost [%]	Travel [mm]	Force change [%]
Follower	Tool center	7.8	20.4	0.7	0.7	48	3.6
Follower	Estimate	7.8	18.9	0.7	0.7	24	3.6
Follower	True contact	7.6	19.3	0.7	0.7	2	3.6
50 N/m	Tool center	9.7	32.2	8.3	0.3	47	9.7
150 N/m	Tool center	13.7	175.9	19.7	1.0	46	26.7
300 N/m	Tool center	22.2	190.1	38.0	2.7	44	48.5
300 N/m	Estimate	24.6	301.7	40.3	27.3	29	44.1
300 N/m	True contact	7.7	20.9	1.7	0.0	2	4.4

Table 12

Added-mass protocol (simulation, 8 mounting positions $\times$ 3 masses $\times$ 4 repetitions $\times$ 3 robot instances). Force errors are medians.

Condition	Trials	Median [mm]	p95 [mm]	CBW [%]	Cov. [%]	$ \Delta\ f\ $ [%]	$\angle\Delta f$ [deg]
Proposed	288	6.3	20.6	0.0	88.2	1.3	0.6
No recovery	288	6.9	18.8	0.7	96.2	1.3	0.6
Random view	288	8.6	27.5	0.0	90.6	1.7	0.9
<i>Proposed, by disc mass</i>							
0.5 kg	96	9.0	22.9	0.0	92.7	1.6	1.0
1.0 kg	96	6.6	15.3	0.0	90.6	1.4	0.6
1.5 kg	96	4.6	13.2	0.0	81.2	1.1	0.5
<i>Proposed, by robot instance (trials admitted by the gate in parentheses)</i>							
A (??/??)	96	13.2	22.9	0.0	67.7	2.1	1.1
B (??/??)	96	6.2	12.4	0.0	96.9	0.7	0.8
C (??/??)	96	3.2	6.5	0.0	100.0	1.1	0.3

and disc stay within the rated 3-kg payload of the FR3. The mounting positions placed the center of mass 99–112 mm from the tool axis and 60–220 mm along it. The disc was attached after Phase 1, so that the payload calibration saw the bracket alone. Each of the 24 conditions was repeated four times, with a remounting tolerance of 0.5 mm and slightly perturbed start configurations, on each of three robot instances, i.e., simulated arms with different model errors drawn from Table 5 (288 trials, Table 12).

With the proposed method, the median error was 6.3 mm and the p95 error 20.6 mm, and the force was measured within a median of 1.3% and 0.6°; 0.0% of the trials were CBW, and the 95% ellipsoid contained the reference point in 88.2% of the trials. The accuracy followed the mass, with median errors of 9.0, 6.6, and 4.6 mm for 0.5, 1.0, and 1.5 kg, as predicted by the $1/\|f\|$ factor of (5).

The traceable reference separates the precision of the sensorless measurement from its trueness. Across the four repetitions of each condition, the estimate scattered by a median of 2.0 mm (precision), and the mean of the repetitions deviated from the reference by a median of 6.1 mm (trueness). The trueness term is set by the static torque error of the individual arm and is the natural target of a richer model calibration.

Applied to this protocol, the deliberately conservative gate of Section 4.6 admitted ?? of the ?? trials (Table 12).

Active selection reduced the p95 error from 27.5 mm with random views to 20.6 mm. The added-mass protocol validates the accuracy of the measurement chain against a traceable reference; the failure mode that recovery prevents is exercised by the diverse contact geometries of Section 4, and the real-robot study validates the two separately (Section 4.9).

4.9. Real-robot validation

Setup and reference The estimator ran on a Franka Research 3 using only q , $\dot{q}$, and the commanded torque of the torque-control interface, reconstructed as the user command plus the gravity torque that the controller adds internally.

Table 13

Uncertainty of the real-robot references (standard uncertainties, GUM).

Component	Evaluation	Standard uncertainty
<i>Equivalent point force (added mass)</i>		
Mounting position on the bracket	Machining tolerance (Type B)	0.1 mm
Bracket pose in the tool frame	Pinned adapter, touch-probe check (Type A)	0.5 mm
Disc center of mass relative to its seat	Thickness and flatness measurement (Type B)	0.2 mm
Remounting repeatability	Repeated touch probing (Type A)	0.3 mm
Disc mass	Calibrated scale (Type B)	0.05%
Base inclination	Inclinometer (Type B)	0.1°
Combined: point / force magnitude / direction	Root sum of squares	0.6 mm / 0.05% / 0.1°
<i>Physical contact (force gauge on a cross-slide)</i>		
Dimple position in the tool frame	Touch probing (Type A)	0.3 mm
Bracket pose in the tool frame	Pinned adapter, touch-probe check (Type A)	0.5 mm
Tip seating in the dimple	Tip and dimple geometry (Type B)	0.3 mm
Force gauge	Calibration certificate (Type B)	0.5%
Gauge axis direction	Inclinometer setting, inertial-sensor log (Type A)	0.4°
Combined: point / force magnitude / direction	Root sum of squares	0.7 mm / 0.5% / 0.4°

The joint-torque sensors of the arm were logged but used only offline, to replay each trial with the sensor-based external-torque estimate of the robot in place of the sensorless residual, which quantifies the cost of sensorless operation. Because the FR3 closes its joint torque loop internally, its actuation is more faithful than that of a current-controlled drive; to emulate the latter, 24 additional trials scaled the commanded torques in software by random per-joint factors of up to $\pm 3\%$ that were unknown to the estimator. The bracket of Section 4.8, machined from aluminum with an oblique arm that carries eight mounting positions (dowel pin and screw), was held in the gripper by a pinned adapter; its geometry was withheld from the estimator. Two apparatus applied the force. Fig. 8 illustrates both apparatus. For the equivalent point force, steel discs of 0.5, 1.0, and 1.5 kg with a radius of 60 mm, weighed on a calibrated scale, were mounted at 45° by an operator after Phase 1 while the arm held its pose; the reference point was the center of mass of the mounted disc in the tool frame, obtained from the drawing of the bracket and the measured disc thickness and verified by touch probing, and the reference force was the weight of the disc. For physical contact, a digital force gauge with a conical steel tip pressed into 1-mm dimples at eight marked points on the faces of the bracket, whose positions in the tool frame were obtained by touch probing. The gauge was mounted on a passive cross-slide, i.e., two low-friction linear guides perpendicular to its axis on a fixed stand, and an operator pushed it along its axis: the slide lets the tip follow the moving contact point during reorientation without lateral stiffness, as Section 4.7 requires, while it keeps the gauge axis, and hence the direction of the pressing force, fixed in the world frame; it realizes the follower pusher of Section 4.7. The axis direction was set with an inclinometer before each trial and logged by an inertial sensor on the gauge, and the gauge reading gave the reference force magnitude. Neither reference entered the estimator. Table 13 lists the uncertainty of the references.

Protocol The validations are separated. Accuracy was validated live with the added mass and the protocol of Section 4.8: 8 mounting positions and 3 masses, each repeated four times (96 trials). Physical contact was validated live with 48 trials: 8 marked points, 2 force levels of about 10 and 20 N, and 3 repetitions; for each point, the pressing direction was set once, within 30° of the surface normal at the initial orientation, and then held fixed in the world frame by the cross-slide. Every trial recorded all five views and their measurement windows; the stopping rule was applied to the logged sequence, which gives the same result as stopping online because the estimator is causal. The effect of recovery was validated by replaying the logged windows of all 144 trials through the estimator with and without recovery, with 20 random seeds of the particle filter per trial; both variants then process identical real data and differ only in the recovery, and the 2880 replays per variant resolve the rare failures that recovery prevents (bootstrap intervals over trials). Random next-view selection changes the physical motion and was therefore run live on 24 added-mass trials, one per mounting position and mass. The Phase-I protocol and all estimator parameters were identical to the simulation study (Table 4).

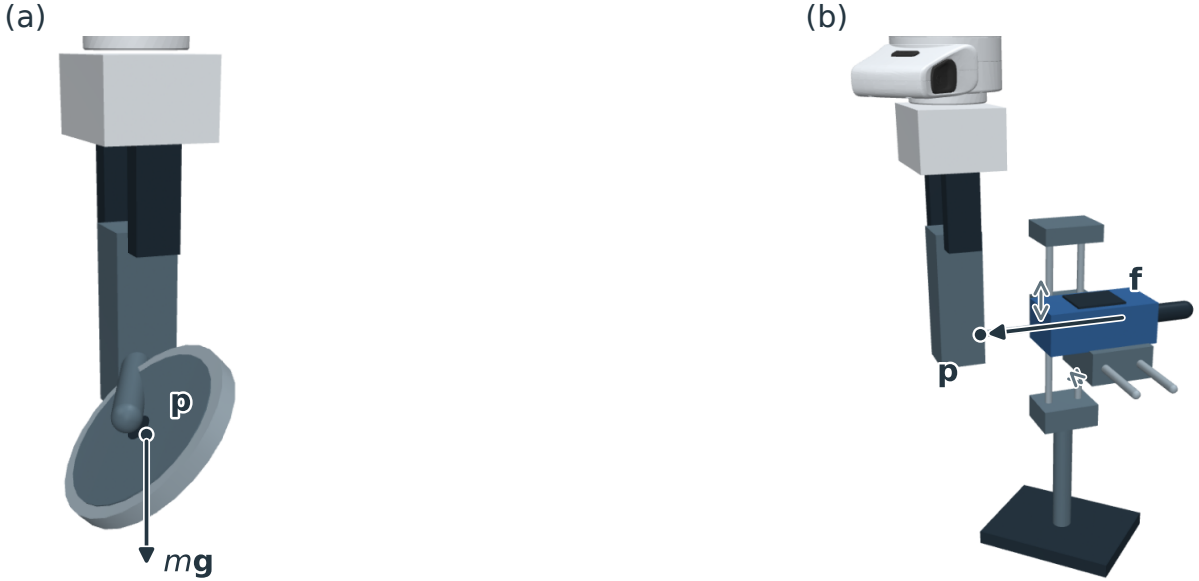


Figure 8: Reference apparatus of the real-robot validation, rendered in MuJoCo. (a) A steel disc is attached after bracket calibration. Its center of mass p and weight mg define the point and force references. The protocol uses eight mounting positions and masses of 0.5, 1.0, and 1.5 kg; a 1-kg disc of radius 60 mm is shown. (b) A force gauge presses at a marked dimple p . Two passive slide directions (double-headed arrows) allow lateral following while fixing the gauge axis in the world frame; the operator supplies the axial force f . The protocol uses eight marked points and approximately 10 and 20 N. Fixture geometry is schematic.

Results Table 14 summarizes the hardware results. The frozen trust gate admitted 126 of the 144 trials; the results below include all trials. With the added mass, the proposed method attained a median position error of HW_{MED} mm and a p95 error of HW_{PNF} mm, and measured the force within a median of $HW_{FMAG\%}$ and HW_{FANG° ; the accuracy followed the mass, with median errors of HW_{MA} , HW_{MB} , and HW_{MC} mm for 0.5, 1.0, and 1.5 kg. Across the four repetitions of each condition, the estimate scattered by a median of HW_{REP} mm (precision), and the mean of the repetitions deviated from the reference by a median of HW_{BIAS} mm (trueness). With the pressed contact, the median error was HW_{HCMED} mm and the p95 error HW_{HCPNF} mm, and the force magnitude matched the gauge within a median of $HW_{HCFMAG\%}$. Between the windows of a trial, the logged gauge axis changed by a median of HW_{HCDIR° and the force magnitude by a median of $HW_{HCFVAR\%}$, which confirms that the force stayed approximately constant in the world frame. Localization required HW_{VIEWS} views and HW_{TLOC} s on average.

The reported uncertainty held on the real robot. None of the 144 live trials was CBW. The 95% particle ball of radius r_{95} contained the reference point in $HW_{BALL\%}$ of the added-mass and $HW_{HCBALL\%}$ of the contact trials. Of the HW_{STOPN} trials that stopped before the fifth view, the reference lay inside the reported r_{95} at the stop in $HW_{STOPBALL\%}$, and $HW_{STOPF20}$ had an error above 20 mm. On identical data, recovery lowered the CBW rate of the replays from $HW_{RNCBW\%}$ to $HW_{RFCBW\%}$ and raised the r_{95} -ball coverage from $HW_{RNBALL\%}$ to $HW_{RFBALL\%}$. With the torque-sensor residual, the median error of the replays was HW_{TSMED} mm, against HW_{RFMED} mm sensorless: dispensing with joint-torque sensing cost HW_{TSCOST} mm of median accuracy. Random views raised the p95 error of the added-mass trials to HW_{RNDPNF} mm. Scaling the commanded torques by up to $\pm 3\%$ raised the median error to HW_{SCMED} mm and lowered the r_{95} -ball coverage to $HW_{SCBALL\%}$, as predicted by the torque-constant study of Table 10.

5. Discussion

What is and is not identifiable The analysis separates two sources of error that are usually conflated. Noise and model error limit the accuracy perpendicular to the line of action and scale as $\sigma_m / \|f\|$; the component along the line is not limited by noise at all but is structurally unobservable from one view. Improving sensors, observers, or filters

Table 14

Real-robot results. **Franka Research 3, bracket with 8 mounting positions or marked points.** Replay: the logged windows of all 144 live trials, processed with 20 filter seeds each. Cov.: fraction of trials in which the 95% credible ellipsoid, or the 95% particle ball of radius r_{95} , contains the reference point.

Variant	Trials	Median [mm]	p95 [mm]	CBW [%]	Cov. ellipsoid [%]	Cov. r_{95} [%]	$ \Delta f $ [%]
<i>Equivalent point force (added mass), live</i>							
Proposed	96	HWMED	HWPNF	0.0	HWCOV	HWBALL	HWFMAG
Random view	24	HWRNDMED	HWRNDPNF	HWRNDCBW	HWRNDCOV	HWRNDBALL	HWRNDFMAG
Commands scaled by $\pm 3\%$	24	HWSCMED	HWSCP NF	HWSCCBW	HWSCCOV	HWSCBALL	HWSCFMAG
<i>Physical contact (force gauge on a cross-slide), live</i>							
Proposed	48	HWHCMED	HWHCPNF	0.0	HWHCCOV	HWHCBALL	HWHCFMAG
<i>Replay of all 144 trials, 20 filter seeds each</i>							
Proposed	2880	HWRFMED	HWRFPNF	HWRFCBW	HWRFCOV	HWRFBALL	HWRFFMAG
No recovery	2880	HWRNMED	HWRNPNF	HWRNCBW	HWRNCOV	HWRNBALL	HWRNFMAG
Torque-sensor residual	2880	HWTSMED	HWTSPNF	HWTSCBW	HWTSCOV	HWTSBALL	HWTSEFMAG

cannot remove the second component; changing the measurement configuration can, and (5) states by how much. The force bins of Table 10 follow the first component, and the angle sweep of Table 6 the second.

Why effective-sample-size recovery fails The effective sample size measures how evenly the weights are spread, not whether the particles span the directions that future measurements will resolve. After the first view, the weight concentrates on the few particles that happen to lie near the line, so the classical rule fires; resampling then copies these particles, and the posterior becomes both shorter along the line and more confident. This is why D1 alone increased the CBW rate. The targeted rules instead ask what the data can and cannot resolve: D3 restores the direction that the data cannot resolve yet, and D2 detects a narrow posterior that contradicts the data. The two failures occur in different trials, which is why each rule alone leaves a higher CBW rate than their combination.

Measurement reliability The CBW rate and the credible-region coverage capture what matters for downstream use of a sensorless measurement: whether the reported uncertainty can be trusted [3]. The uncertainty budget is what makes the particle posterior usable as a measurement result. Its Type B terms describe effects that contact-free calibration cannot see, and when the true torque-constant error or follower variation exceeds the values assumed in the budget, the coverage drops (Table 10). On a current-controlled arm, u_k should therefore be taken from the tolerance of the drive's torque constant, and ε_f from the specification of the pusher or the expected variability of a human hand.

Recursive and batch estimation The profile likelihood can also be maximized without recursion. On identical measurement windows, a batch re-estimate with a rank-aware covariance reached a similar final accuracy and a better-calibrated reported uncertainty than the particle filter (Supplementary material, Section S1). The recursive filter is what provides the posterior between views on which recovery and view selection act, so the two are complementary, and a batch re-estimate can serve as the final report of an episode.

Limitations The method assumes a single follower contact at a fixed material point with an approximately constant world-frame force during an episode. Sliding contacts, contact switching, multiple simultaneous contacts, and contacts with a fixed environment, whose force direction changes with the reorientation, violate these assumptions and are outside the present scope. The pusher must be compliant tangentially to the surface, because a laterally stiff pusher turns the motion of the contact into force changes between views. Low forces remain the hardest regime, as predicted by the $1/\|f\|$ scaling, and the motion is confined to the envelope validated in Phase 1. The reported uncertainty is optimistic in the tail (81.9% coverage of the nominal 95% ellipsoid); a certified bound requires an independent calibration of the reported region. The information criterion is a local, fixed-covariance approximation; strongly multimodal posteriors may benefit from a non-myopic objective.

6. Conclusion

We presented a sensorless measurement of contact location and force on an unknown grasped object that uses only joint states and commanded torques. We showed that the single-view measurement is structurally ambiguous along the line of action, that two transverse views make it identifiable, and how the attainable accuracy scales with force magnitude and view separation. The estimator combines a validated residual with a GUM-type uncertainty budget and a force-profiled particle filter in which every step targets the same posterior; it preserves the physically valid ambiguity, detects when its support becomes unreliable, and actively changes the measurement configuration until the contact is observable. In 1,200 paired simulation trials with realistic model and actuation errors, the method reduced the 95th-percentile error by a factor of ?? relative to a regularized particle filter and the confident-but-wrong rate from ??% to 0.8%, and the actively selected second view reached a 95th-percentile error of 25.9 mm, against 40.4 mm for a random second view. **On the real robot, the method localized the equivalent point force of added masses with a median error of HWMED mm against a traceable reference and pressed contacts with HWHCMED mm, and its reported radius remained reliable.**

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The simulation code, the per-trial results that support the findings, and supplementary material are available on the project page, <https://anonymous.4open.science/w/apcl/>.

A. No-data invariance of the recovery

A recovery rule may change where the particles are, but it should not change the posterior that they approximate. We tested this property directly. From the simulated windows of 10 development trials, we built filter states after one view and after two views, applied each rule (D1, D2, and D3; Section 3.5) ten times in succession without any new data, and compared the particle summary with an independent importance sample of the same posterior (300,000 draws from an equal mixture of a uniform proposal over $\mathcal{B}$ and a Gaussian proposal three times wider than the filter posterior, weighted by target over proposal density). Table A.1 reports the ratio of r_{95} , and of the spread perpendicular to the line of action, to the reference, pooled over the three rules.

The ratios remain at one within the Monte Carlo error of repeated resampling, however often a rule is applied, and show no drift. Before any recovery, the filter states themselves matched the reference (median r_{95} ratios of 1.03 and 0.98 after one and two views), as expected from the posterior-invariant Metropolis–Hastings move. The test is sensitive to a re-supply that does not preserve the target: a re-supply that resamples, injects the same proposals, and then reweights all particles by the full accumulated likelihood counts the data a second time, because resampled particles already represent the posterior, and ten such applications shrank r_{95} after two views to 0.33 of the reference and the perpendicular spread after one view to 0.26.

Table A.1

Recovery applied repeatedly without new data (10 filter states per row; ratios to an independent reference sample of the same posterior; median and range over states and rules).

State	Reference r_{95} [mm]	r_{95} ratio, 1 application	r_{95} ratio, 10 applications	Perpendicular-spread ratio, 10 applications
After one view	176.9	1.00	1.00 (0.97–1.01)	0.99 (0.87–1.10)
After two views	20.8	1.00	0.99 (0.87–1.27)	0.99 (0.88–1.54)

B. Supplementary data

Supplementary material to this article is provided as a separate file. Section S1 compares the recursive estimator with a batch estimator on identical measurement windows, and Section S2 reports calibrated reporting with false-report rates per trial and per report.

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